

Inflammatory Myopathy — Workup & Treatment

1st LINE

PRED
0.75~1.5 mg/kg

3-6 months

goal

1g IV x 3

goal

≤10 mg

taper

5 mg q2-4wk

FASTEST onset

MTX

7.5-25 mg/wk

AVOID in ILD!

FASTEST onset

AZA

2-3 mg/kg

TPMT check

SLOW

6-18 mo

MMF

1g BID → 3g

DRUG OF CHOICE for ILD

IVIG

2 g/kg over 2-5 days

monthly x ≥3

FDA approved DM

HMCCR MONOTHERAPY

RTX

750 mg/m² q6-18mo

refractory/ILD

RELAPSE

CK rising

STERIOD MYOPATHY

BIG ERROR

DM	✓	✓	✓	✓	IVIG FDA
PM	✓	✓	sometimes	✓	✓
ASyS			MMF	RTX	early if ILD
IMNM	⚠	often insufficient	IVIG	YES HMCCR	RTX often
IBM	✗	✗	✗	✗	✗

HMCCR → IVIG alone first

aggressive triple

SRP → MMF + early RTX

ASyS ILD → DO NOT delay for SRP

ONLY supportive

RELAPSE

weakness during taper

CK rising

STERIOD MYOPATHY

weakness on high dose

CK normal

no EMG irritability

DM

IBM

ASyS

IMNM

CPI MYOSITIS

ILD

statin

very high

ONLY supportive

ONLY supportive

NOT cancer progression = IMMUNE TOXICITY

scattered necrotic fibers

MHC-I ↑

BEOP

ECG

ECG

ECG

MYOCARDITIS

1-2 MG-LIKE

MYOSITIS

Arms, weak and can't lift them.

TRIAD!

MYOCARDITIS

STEP 1

STOP DRUG FIRST

STEP 2

IV METHYLPRED

re-challenge generally AVOIDED

STEP 3

IVIG

ADD if severe MG features

poor steroid response

re-challenge AVOIDED

1. Prednisone 1st line

WHAT YOU SEE

Moon/sun icons labeled PRED 0.75-1.5 mg/kg first line

WHAT IT ENCODES

High-dose oral corticosteroids (prednisone ~0.75–1.5 mg/kg/day, commonly ~1 mg/kg) are FIRST-LINE for DM, PM, IMNM, and antisynthetase syndrome. Steroids are continued at the high dose until strength improves and CK normalizes, then SLOWLY tapered over months. A steroid-sparing agent is added early (often at the outset) to limit cumulative steroid toxicity. IBM is the exception — it does not respond.

BOARD TRAP

WRONG answer: tapering prednisone rapidly once the CK drops. Discriminator: too-fast tapering causes relapse; the taper must be gradual over months and guided by clinical strength, with a steroid-sparing agent on board to permit dose reduction.

2. IV methylpred pulse

WHAT YOU SEE

Sign reading 1g IV x3 then taper

WHAT IT ENCODES

Severe or rapidly progressive disease (profound weakness, dysphagia/aspiration risk, myocarditis, or rapidly progressive ILD) begins with PULSE IV methylprednisolone — e.g., 1 g IV daily for 3 days — before transitioning to high-dose oral prednisone. Pulse dosing achieves rapid immunosuppression in organ- or life-threatening presentations.

BOARD TRAP

WRONG answer: starting a patient with severe dysphagia and respiratory compromise on standard oral prednisone only. Discriminator: severe/fulminant disease warrants IV pulse methylprednisolone (± early IVIG) up front — relying on oral steroids alone delays control in a patient who is aspirating or in respiratory failure.

3. Treat strength not CK

WHAT YOU SEE

Box reading TREAT STRENGTH NOT CK

WHAT IT ENCODES

Therapy is titrated to CLINICAL STRENGTH (and function), not to the CK number. CK may lag, rise transiently with exercise, or stay up while the patient is clearly improving; conversely a falling CK with worsening strength suggests steroid myopathy or a non-inflammatory process. Track manual muscle testing, functional tasks, and patient-reported activity alongside the enzymes.

BOARD TRAP

WRONG answer: escalating immunosuppression solely because the CK ticked up while the patient is getting stronger. Discriminator: an isolated CK rise without clinical worsening does not warrant more immunosuppression — treat the patient's strength, not the lab; over-treating off the CK adds toxicity.

4. Methotrexate

WHAT YOU SEE

MTX bottle on the shelf, fastest onset

WHAT IT ENCODES

Methotrexate is a first-line steroid-sparing agent with relatively FAST onset, given weekly (with folic acid). It is effective for DM/PM muscle and skin disease. Key caveat: AVOID in significant interstitial lung disease because of methotrexate pneumonitis/fibrosis risk that can be confused with or worsen ILD.

BOARD TRAP

WRONG answer: choosing methotrexate as the steroid-sparing drug in an antisynthetase/anti-Jo-1 patient with ILD. Discriminator: with significant ILD, MTX is contraindicated/avoided (pulmonary toxicity) — use mycophenolate or rituximab instead; the lung status decides the agent.

5. Azathioprine

WHAT YOU SEE

AZA bottle, slow onset, check TPMT

WHAT IT ENCODES

Azathioprine is a steroid-sparing alternative to methotrexate with SLOWER onset (months to full effect). Check TPMT (thiopurine methyltransferase) activity BEFORE starting — low/absent TPMT predicts severe myelosuppression — and monitor CBC and LFTs. It is a reasonable choice when MTX is contraindicated (e.g., mild ILD, liver disease, alcohol use).

BOARD TRAP

WRONG answer: starting full-dose azathioprine without checking TPMT. Discriminator: TPMT-deficient patients develop life-threatening bone-marrow suppression on standard azathioprine doses — TPMT testing (or genotyping) before initiation, plus CBC monitoring, is mandatory.

6. MMF drug of choice for ILD

WHAT YOU SEE

MMF bottle labeled drug of choice for ILD

WHAT IT ENCODES

Mycophenolate mofetil (MMF) is the steroid-sparing agent of choice when INTERSTITIAL LUNG DISEASE accompanies myositis (antisynthetase syndrome, anti-MDA5 DM). It treats both muscle and lung, is generally well tolerated (GI upset, cytopenias), and is teratogenic so requires contraception. Rituximab is the main alternative/add-on for ILD-predominant or refractory disease.

BOARD TRAP

WRONG answer: defaulting to methotrexate for steroid-sparing in myositis-ILD. Discriminator: MMF (or rituximab) is preferred when ILD is present because MTX risks pulmonary toxicity — match the agent to the lung disease.

7. IVIG

WHAT YOU SEE

IV bag labeled IVIG, fast onset

WHAT IT ENCODES

IVIG has RAPID onset and is FDA-approved for dermatomyositis; it is especially valuable in DM, IMNM (notably anti-HMGCR), severe dysphagia, and when infection risk limits other immunosuppression (e.g., active infection, pregnancy). Given as a course every few weeks; watch for headache/aseptic meningitis, thrombosis, and renal effects.

BOARD TRAP

WRONG answer: viewing IVIG only as a last-line agent for refractory disease. Discriminator: IVIG is an early/first-line option in DM and anti-HMGCR IMNM (and for life-threatening dysphagia) because of its speed and favorable infection profile — not merely a salvage therapy.

8. Rituximab

WHAT YOU SEE

The 'RTX' drug bottle on the lower shelf among the steroid-sparing agents — rituximab, the B-cell-depleting option reserved for refractory DM/PM/ASyS/IMNM (but useless in IBM).

WHAT IT ENCODES

Rituximab (anti-CD20 B-cell depletion) is used for REFRACTORY myositis and is favored in antisynthetase syndrome (especially ILD-predominant) and anti-SRP/anti-HMGCR necrotizing myopathy. Typical dosing mirrors other rheumatic indications (e.g., two 1 g infusions 2 weeks apart, or weekly x4). Screen for hepatitis B and check immunoglobulins before use.

BOARD TRAP

WRONG answer: reserving rituximab indiscriminately for all refractory cases including IBM. Discriminator: rituximab helps refractory DM/PM/antisynthetase/IMNM but does NOT work for IBM, which is immunotherapy-resistant — escalating to RTX in IBM exposes the patient to risk without benefit.

9. Steroid myopathy

WHAT YOU SEE

Sign reading steroid myopathy, weakness with normal CK

WHAT IT ENCODES

Steroid (glucocorticoid) myopathy is a common iatrogenic MIMIC of disease relapse: new proximal weakness developing while on high-dose or prolonged steroids, but with a NORMAL CK and a NON-irritable (myopathic without spontaneous activity) EMG and no inflammation on biopsy. It is dose- and duration-related and improves when steroids are reduced.

BOARD TRAP

WRONG answer: increasing the steroid dose for new weakness in a patient on chronic high-dose prednisone. Discriminator: in true myositis flare the CK rises and EMG is irritable; in steroid myopathy the CK stays NORMAL and EMG is non-irritable — the correct move is to TAPER the steroid, not escalate it.

10. Treatment response table

WHAT YOU SEE

Grid of checkmarks/X for DM, PM, ASyS, IMNM, IBM

WHAT IT ENCODES

Response to immunosuppression by subtype: DM, PM, and antisynthetase syndrome generally RESPOND; IMNM (especially anti-HMGCR) responds, often needing IVIG/rituximab, while anti-SRP can be refractory; IBM is essentially NON-responsive (X across the board). This table is the high-yield 'who gets immunosuppression' summary.

BOARD TRAP

WRONG answer: applying the standard steroid + steroid-sparing immunosuppression algorithm uniformly to all five subtypes including IBM. Discriminator: IBM is the exception — it does not meaningfully respond to immunosuppression, so the algorithm that works for DM/PM/ASyS/IMNM does not apply to it.

11. IBM only supportive

WHAT YOU SEE

Elderly figures with cane labeled only supportive

WHAT IT ENCODES

Inclusion body myositis: typically a man over 50, with DISTAL and ASYMMETRIC weakness (classically finger flexors and quadriceps), frequent dysphagia, slowly progressive course, near-normal CK, and rimmed vacuoles with p62/TDP-43 inclusions on biopsy. There is no effective immunotherapy — management is SUPPORTIVE (PT/OT, assistive devices, dysphagia/aspiration precautions, falls prevention).

BOARD TRAP

WRONG answer: trialing high-dose steroids and steroid-sparing agents in confirmed IBM hoping for response. Discriminator: IBM is steroid- and immunotherapy-REFRACTORY; immunosuppression adds toxicity (including steroid myopathy) without benefit — supportive care is the standard of care.

12. Myocarditis / cardiac

WHAT YOU SEE

ECG and heart alarm panel on the right

WHAT IT ENCODES

Cardiac involvement (myocarditis, conduction disease, arrhythmia, rarely heart failure) is a feared and prognosis-worsening complication of inflammatory myopathy, more common with anti-SRP disease and in checkpoint-inhibitor myositis. Screen with ECG and troponin (note troponin-T can rise from skeletal muscle — use troponin-I to confirm cardiac source), with echo/cardiac MRI as needed.

BOARD TRAP

WRONG answer: attributing an elevated troponin-T purely to cardiac injury in a myositis patient. Discriminator: skeletal muscle releases troponin-T, so an isolated troponin-T rise may be muscular — confirm cardiac involvement with the more cardiac-specific troponin-I plus ECG and imaging before diagnosing myocarditis.

13. ICI myositis TRIAD

WHAT YOU SEE

Yellow triangle labeled TRIAD with myositis, myocarditis, MG

WHAT IT ENCODES

Immune-checkpoint-inhibitor (ICI) myositis is an immune-related adverse event that classically presents as an OVERLAP TRIAD of myositis + MYOCARDITIS + myasthenia-gravis-like neuromuscular disease (ptosis, diplopia, bulbar/respiratory weakness). The myocarditis component carries high mortality, so any ICI patient with weakness needs urgent cardiac evaluation (ECG, troponin, echo/MRI).

BOARD TRAP

WRONG answer: treating ICI-related weakness as isolated myositis and skipping cardiac and neuromuscular assessment. Discriminator: ICI myositis overlaps with myocarditis and MG — missing the myocarditis (the lethal component) by not checking ECG/troponin is the dangerous error; evaluate the whole triad.

14. Stop ICI first

WHAT YOU SEE

Red STOP sign reading stop drug first

WHAT IT ENCODES

Management of ICI-induced myositis/myocarditis: STEP 1 — STOP the checkpoint inhibitor immediately; STEP 2 — high-dose corticosteroids (IV methylprednisolone for severe/cardiac disease) ± IVIG/plasmapheresis. Because severe irAEs (especially myocarditis) can be fatal and recur, RE-CHALLENGE with the ICI is generally avoided after high-grade myositis/myocarditis.

BOARD TRAP

WRONG answer: re-starting the checkpoint inhibitor once the myositis/myocarditis settles because the tumor was responding. Discriminator: after severe (grade 3–4) ICI myositis or any myocarditis, re-challenge is generally CONTRAINDICATED due to risk of fatal recurrence — permanent discontinuation plus immunosuppression is the rule.

15. EMG irritable myopathy

WHAT YOU SEE

The diagnostic-ladder tier near the top-center where the EMG/workup steps stack toward 'DM / PM / IBM / ASyS / myositis' — EMG sits in this work-up rung, showing an irritable myopathic pattern that confirms myopathy and guides the biopsy site.

WHAT IT ENCODES

NEW element. Electromyography in active inflammatory myopathy shows an IRRITABLE MYOPATHIC pattern: short-duration, low-amplitude, polyphasic motor unit potentials PLUS spontaneous activity (fibrillations, positive sharp waves, complex repetitive discharges). EMG confirms a myopathic (not neuropathic) process, gauges activity, and helps localize a biopsy site; MRI (STIR edema) complements it for site selection.

BOARD TRAP

WRONG answer: interpreting a normal or non-irritable EMG as excluding myositis, or reading large/long motor units as confirming it. Discriminator: long-duration, high-amplitude units indicate a NEUROPATHIC process (e.g., ALS, neuropathy), not myopathy; and a patchy myopathy can spare the sampled muscle — a non-diagnostic EMG does not rule it out.

16. Hypothyroid myopathy mimic

WHAT YOU SEE

The workup branch (mid-panel diagnostic ladder) routing the high-CK/proximal-weakness presentation through non-inflammatory mimics before landing on myositis — hypothyroid myopathy is the mimic to exclude here (check TSH; a high CK can come from low thyroid, not inflammation).

WHAT IT ENCODES

NEW element. Hypothyroid myopathy is a key non-inflammatory MIMIC: proximal weakness, myalgia, cramps, slowed/delayed deep-tendon reflex relaxation ('hung-up' reflexes), and an ELEVATED CK — sometimes markedly — that can falsely suggest myositis. The fix is thyroid hormone replacement, not immunosuppression. Always check TSH in a new myopathy workup before committing to inflammatory myositis.

BOARD TRAP

WRONG answer: starting corticosteroids for proximal weakness and a high CK without checking thyroid function. Discriminator: hypothyroidism raises CK and causes proximal weakness with hung-up reflexes; a high TSH points to thyroid-replacement therapy, not steroids — missing the TSH is the classic avoidable error.

17. PMR vs myositis

WHAT YOU SEE

The workup branch separating true myositis from the older-adult girdle-pain distractor — PMR sits here as the look-alike: pain/stiffness with NORMAL CK and intact strength, the opposite of the high-CK true-weakness myopathies.

WHAT IT ENCODES

NEW element. Polymyalgia rheumatica (PMR) is the classic distractor: age >50 with proximal shoulder/hip-girdle PAIN and STIFFNESS (worse in the morning), high ESR/CRP, but a NORMAL CK and no true weakness (limited by pain, not power). It responds dramatically to LOW-dose prednisone (~15 mg/day) and may be associated with giant cell arteritis — quite different from the high-CK, true-weakness, high-dose-steroid inflammatory myopathies.

BOARD TRAP

WRONG answer: diagnosing polymyositis in an older patient with girdle pain, stiffness, and a high ESR. Discriminator: PMR has PAIN/STIFFNESS with NORMAL CK and intact strength and responds to LOW-dose steroids; polymyositis has true WEAKNESS with HIGH CK needing HIGH-dose steroids plus immunosuppression — the CK and the strength exam separate them.